Co2 - Assessment Tool 2

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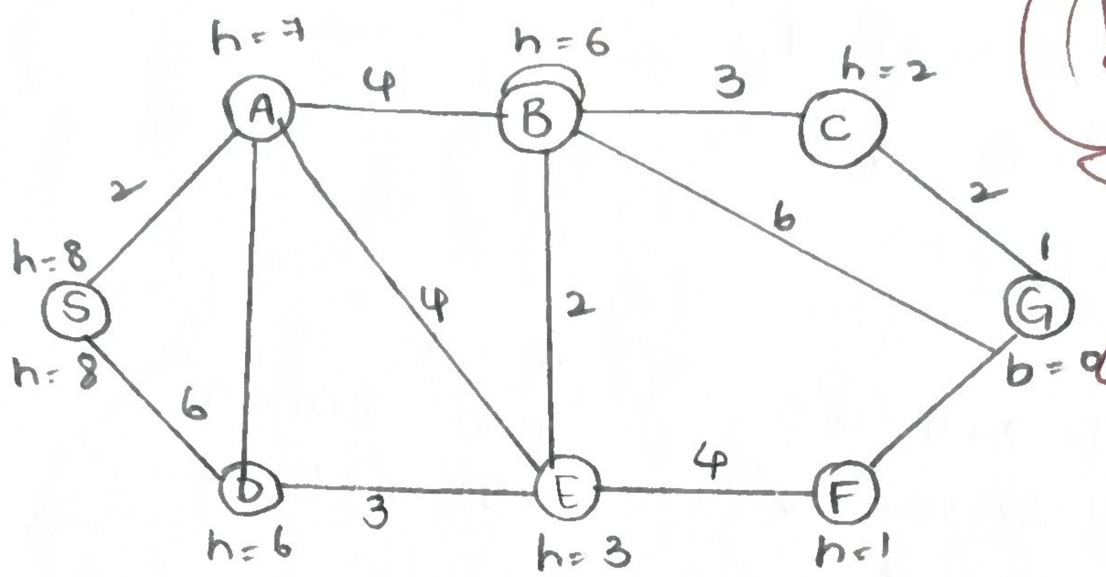
MLA0101 - Artificial Intelligence and Expert Systems.

1. Smart Ambulance Navigation :



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6/8h6

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Let,

$$\boxed{f(n) = g(n) + h(n)}$$

where, $g(n)$ = actual cost from start S to node

$h(n)$ = heuristic estimated cost from node to Goal G

$f(n)$ = total evaluation value.

Heuristic values

Node	h(n)
S	8
A	7
B	6
C	2
D	6
E	3
F	1
G	0.

A^* Exploration

Start at S :

$g(A) = 2,\ h(A) = 7$

$f(A) = 2 + 7 = 9$

From A :

$A \rightarrow E$

$g(E) = 2 + 4 = 6$

$f(E) = 6 + 3 = 9.$

$A \rightarrow B$

$g(B) = 2 + 4 = 6$

$f(B) = 6 + 6 = 12$

From E :

$E \rightarrow F$

$g(F) = 6 + 4 = 10$

$f(F) = 10 + 1 = 11$

From F :

$F \rightarrow G$

Selected Route :

$S \rightarrow A \rightarrow E \rightarrow F \rightarrow G$

Cost : $2 + 4 + 4 + 1 = 11$

$\therefore$ Optimal route $= S \rightarrow A \rightarrow E \rightarrow F \rightarrow G$

Total path cost $= 11.$

Autonomous warehouse Robot :

5	S	?	?	?	?	?	?
4	?	#	?	?	#	?	?
3	?	?	?	#	?	?	#
2	?	#	?	?	?	#	?
1	?	?	#	?	?	?	G
	1	2	3	4	5	6	7

Grid coordinates are :

- Start S = (5,1)
- Goal G = (1,7)

Obstacles shown in grid :

- (4,4)
- (3,7)
- (2,2)
- (2,6)
- (1,3)

Robot's movement

Start : S (5,1)

⇒ Moves towards goal while checking nearby cells.

Route :

(5, 1) → (5, 2) → (5, 3) → (5, 4) → (5, 5) → (5, 6) → (5, 7)

(5, 7) → (4, 7) → (3, 7) X

Blocked, so change root.

(4,7) → (4,6) → (4,5) → (3,5) → (2,5) → (1,5) → ...

(1,7) = Goal state

∴ Final Route :

S → (5,2) → (5,3) → (5,4) → (5,5) → (5,6) → (5,7)
↓
(4,5) ← (4,6) ← (4,7)
↓
(3,5)
↓
(2,5)
↓
(1,5) → (1,6) → (1,7) $\boxed{\text{Goal}}$

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3) University Examination Scheduling

Examination hall	capacity
Hall A	120 students
Hall B	90 students
Hall C	60 students

Examination	E_1	E_2	E_3	E_4	E_5	E_6
No. of Students	100	80	70	60	90	50

Given

exams :

$E_1, E_2, E_3, E_4, E_5, E_6$.

Domain : $\{T_1, T_2, T_3, T_4\}$

Constraints : $E_1 \neq E_2$, $E_1 \neq E_3$,

$E_2 \neq E_4$, $E_3 \neq E_5$,

$E_4 \neq E_6$, $E_2 \neq E_5$

Compute degree

$deg(E_1) = 2$, $deg(E_2) = 3$

$deg(E_3) = 2$, $deg(E_4) = 2$

$deg(E_5) = 2$, $deg(E_6) = 1$

max degree = $deg(E_2) = 3 \rightarrow E_2$.

$\Rightarrow E_2 = T_1$

$D(E_2) = \{T_1\}$

Forward checking

Neighbor = $\{E_1, E_4, E_5\}$

$D(E_1) = \{T_1, T_2, T_3, T_4\} - \{T_1\} = \{T_2, T_3, T_4\}$

$D(E_4) = \{T_2, T_3, T_4\}$, $D(E_5) = \{T_2, T_3, T_4\}$

$\Rightarrow$ Next variable

$E_1 = T_2 \rightarrow D(E_1) = \{T_2\}$

forward check from E_1

$D(E_3) = \{T_1, T_2, T_3, T_4\} - \{T_2\} = \{T_1, T_3, T_4\}$

⇒ Assign $E_4 = T_2$

$T_2 \neq T_1$ ✓ valid

$D(E_4) = \{T_2\}$

forward check from E_4

Neighbor: E_6

$D(E_6) = \{T_1, T_3, T_4\}$

⇒ Assign $E_3 = T_1$

Neighbors = $E_1(T_2), E_5(T_2) \rightarrow T_1 \neq T_2$ ✓ valid

$D(E_3) = \{T_1\}$

⇒ Assign $E_6 = T_1$

Neighbor is $E_4(T_2) \rightarrow T_1 \neq T_2$ ✓ valid

$D(E_6) = \{T_1\}$.

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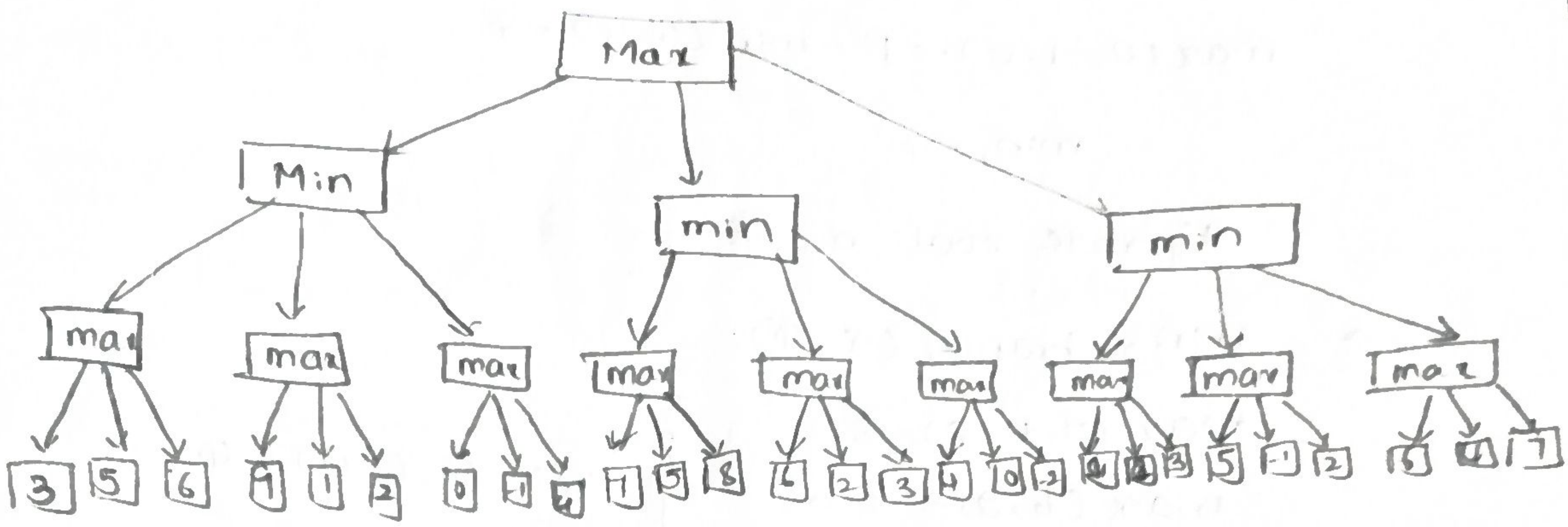
Verification:-

constraint	Value	satisfied
$E_1 \neq E_2$	$T_2 \neq T_1$	✓
$E_1 \neq E_3$	$T_2 \neq T_1$	✓
$E_2 \neq E_4$	$T_1 \neq T_2$	✓
$E_3 \neq E_5$	$T_1 \neq T_2$	✓
$E_4 \neq E_6$	$T_2 \neq T_1$	✓
$E_2 \neq E_5$	$T_1 \neq T_2$	✓

$\therefore T_1 = \{E_2, E_3, E_6\}, T_2 = \{E_1, E_5, E_4\}, T_3 = \emptyset, T_4 = \emptyset.$

we want to approve
Person

Game-playing AI:



Given that

max nodes : (6, 9, 4), (8, 6, 9), (3, 5, 7)

min nodes : min → 4, 4, 3

Root = max (4, 4, 3) = 4.

Pruning : min₁ → leaf 9 ≥ (3, 6)

prune (1, 2)

min 3 → max $_{31}$ = 3 ≤ α (4) → prune

max$_{32}$, max$_{33}$

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Alpha-Beta Pruning (α = -∞)

β = +∞ at root

→ min₁ branch:

max (3, 5, 6) = 6 → β = 6.

max$_{12}$: leaf = 9 → α = 9 ≥ β = 6 → prune (1, 2)

$B(min_1) = min(6, 9) = 6$

$max(0, -1, 4) \rightarrow B = min(6, 4) = 4$

$min_1 = 4$

Update root $\alpha = 4$

$\rightarrow$ min_2 branch $(\alpha = 4)$:

$max(7, 5, 8) = 8$:
$max(6, 2, 3) = 6$
$max(4, 0, -2) = 4$

$\Big\}$ $min(8, 6, 4) = 4.$

$\rightarrow$ min_3 branch $(\alpha = 4)$:

$max(1, 2, 3) = 3 \rightarrow B(min_3) = 3$

$\boxed{Root = max(4, 4, 3) = 4.}$

5. Delivery Route Optimization.

	D	A	B	C	E
D	-	6	9	8	7
A	6	-	3	5	6
B	9	3	-	4	5
C	8	5	4	-	3
E	7	6	5	3	-

Neighbor cost set:

$f(S_1) = 26$, $f(S_2) = 30$, $f(S_3) = 24$, $f(S_4) = 27$
$f(S_5) = 23$, $f(S_6) = 29$

selection rule:

$s^* = \arg\min(s_i) \rightarrow f(s_5) = 23$

$\min\{26, 30, 24, 27, 23, 29\}.$

Move condition:

$f(s_5) = 23 < f(\text{current}) \rightarrow$ accept $\rightarrow$ current $= s_5$

Second iteration

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Best neighbor of s_5 : $f = 22$

since $22 < 23 \rightarrow$ accept

current = 22 KM

Termination Test

$f(s_j) < 22$ (all neighbors ≥ 22) $\rightarrow$ local optimality

conditional satisfied $\rightarrow$ stop

$\boxed{f(\text{final}) = 22 \text{ km}}$

∴ Hill climbing guarantees $f(\text{final}) \leq f(s_1)$

but not $f(\text{final}) = f(\text{global minimum}) \rightarrow$ global

optimality not proven.